Supplementary material

Gains on Synthetic Occlusion Are Not Enough: Occlusion Augmentation and Real Occlusion in Person Detection

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Every table in this supplement is generated by `shortpaper5/suplemento_tablas.py` from the project's result files (`informes/analisis_final/`); no figure is typed by hand. Analyses marked as post hoc were declared after seeing the results and decide no hypothesis.

1 Objectives, method and evidence

Table S1: Mapping of specific objectives, method, evidence and pending work.

Objective	Method	Evidence	Pending
OE1. Degradation under occlusion	Stratified AP@0.5 with *ignore* regions, three sources	0.82 (unoccluded) to 0.24 (severe), YOLOv11s on ExDark	—
OE2. Effect of augmentation by occluder	V3 vs. V1, control and 3 real sources; patches, silhouettes, diffusion, real geometry and depth	17 of 17 controls; primary count (equal exposure, 33 cells): 5 pass, 15 rule out +2 pp and 13 are inconclusive; mean +0.96 on CityPersons, +0.01 on NightOwls and −0.94 on ExDark	Occlusion mined from real images
OE3. Dependence on the detector	YOLOv11s, YOLOv11m, YOLO26s, RT-DETR-l, RetinaNet, Faster R-CNN	Same pattern in the first five; Faster R-CNN transfers with silhouettes on NightOwls and CityPersons (H15 refuted), but not with diffusion or depth (H16a and H16b refuted), and RetinaNet with silhouettes does not transfer (H17)	Another two-stage detector or backbone
OE4. Domain and data volume	Retraining on CityPersons (H6, H11); NightOwls from 2,000 to 20,000 images (H7)	The deficit persists; the in-domain gain was not confirmed (H11)	—
OE5. Mechanism	Recall, false positives, confidence; visibility; realism (H13); cross-family matrix	Synthetically occluded people remain distinguishable from real ones; lower recall with higher confidence	Confirmatory test of the mechanism
OE6–OE7. Attention module; annotation reliability	FPS and parameters; inter-annotator κ	No gain from the module (H1, H2), FPS at 82 %; κ 0.06–0.60	Embedded-device test

Variants of the original design. Experiment 1 had four variants: V1 (baseline), V2 (CBAM attention module in the neck), V3 (occlusion augmentation) and V4 (both). H1 (V4 improves on V1) and H2 (V3 improves more than V2) were not confirmed, and H5 (silhouettes reduce the gap) only partially, because the gap closed by real occlusion getting worse. The attention module cut FPS to 82 % (161–184 versus 144–157) with no gain on real occlusion. From iteration 2 on the question is transfer and only V1 and V3 are compared.

2 Hypotheses and pre-specification

The global SHA-256 fingerprint of the protocol proves that no frozen file changed after the freeze, but the date is recorded by the project itself (`CAMBIOS_PROTOCOLO.md`), not by a third party. The "first run" column comes from each run's log and shows that every freeze precedes training. All dates are in 2026.

Table S2: Hypotheses, prediction, freeze and verdict. The fingerprint is that of the frozen protocol (global SHA-256, first 8 characters); the date is self-recorded by the project, not by a third party. First run: start of the first valid run of the block, from its log (local time, UTC−5, on the laptop; UTC on RunPod: H11, H12 and H14–H17).

Hyp.	Prediction	File	Version and fingerprint	Frozen	First run	Verdict
H1–H3	V4 and V3 improve occluded AP by ≥2 pp over V1 (H1, H2) and the gain transfers to real occlusion (H3)	`PROTOCOLO.md`	v1.0 `c5002ea5`	2026-09-11 08:09	2026-09-11 08:47	Not confirmed (0 of 3 sources in H3)
H4, H5	With silhouettes the gain transfers (H4); smaller synthetic gain and gap (H5)	`PROTOCOLO_v2.md`	v2.0 `9915d90f`	2026-09-12 12:51	2026-09-12 12:54	H4 not confirmed (0 of 3); H5 partial
H6	No transfer when retraining on CityPersons	`PREREGISTRO_H6_H7.md`	v3.0 `e971ab8b`	2026-09-19 16:48	2026-09-19 16:48	Not detected (0 of 3)
H7	No transfer with 20,000 night images	`PREREGISTRO_H6_H7.md`	v3.1 `b0043c89`	2026-09-25 20:58	2026-09-25 20:59	Not detected (no control)
H8	No transfer with RT-DETR-l	`PREREGISTRO_H8.md`	v3.2 `51c0a2d8`	2026-09-26 18:52	2026-09-26 18:52	Not detected (0 of 3)
H9	No transfer with diffusion occluders	`PREREGISTRO_H9.md`	v3.3b `243acffb`	2026-10-03 01:29	2026-10-03 01:29	Not detected (1 of 3)
H9b	Same with 30 fixed epochs	`PREREGISTRO_H9b.md`	v3.4 `51786d9d`	2026-10-06 09:01	2026-10-06 09:02	Not detected (0 of 3)
H10	No transfer with real-occlusion geometry	`PREREGISTRO_H10.md`	v3.5 `1489961a`	2026-10-06 18:27	2026-10-07 05:03	Not detected (1 of 3)
H11	Gain ≥2 pp at CityPersons visibility 0.20–0.50 with equal exposure	`PREREGISTRO_H11.md`	v3.6b `bd2666fd`	2026-10-13 14:38	2026-10-13 20:37	Not confirmed
H12	No transfer with YOLOv11m (silhouettes, H12a; diffusion, H12b)	`PREREGISTRO_H12.md`	v3.7b `ef6138c1`	2026-10-13 21:06	2026-10-17 14:50	Not detected (0 of 3 and 0 of 3)
H13	AUC(patches) > AUC(silhouettes) > AUC(diffusion) with DINOv2	`PREREGISTRO_H13.md`	v3.8 `7ce00c9c`	2026-10-17 09:48	no training	Partially confirmed
H14	No transfer with depth order	`PREREGISTRO_H14.md`	v3.9 `264ac449`	2026-10-17 10:10	2026-10-17 16:33	Not detected (1 of 3)
H15	No transfer with Faster R-CNN	`PREREGISTRO_H15.md`	v3.10 `b7843b3a`	2026-10-20 16:20	2026-10-21 13:31	Passes the rule (2 of 3)
H16a, H16b	With Faster R-CNN transfer also occurs with diffusion (H16a) and depth order (H16b)	`PREREGISTRO_H16_H17.md`	v3.11 `e9b85aee`	2026-10-22 07:50	2026-10-22 13:57	Refuted (0 of 3 and 0 of 3)
H17	No transfer with RetinaNet (one-stage, same backbone as H15, its own reference recipe)	`PREREGISTRO_H16_H17.md`	v3.11 `e9b85aee`	2026-10-22 07:50	2026-10-22 13:08	Not detected (0 of 3)

3 Run log and recipes

The valid runs are: 14 in experiment 1 (12 of V1–V4 and two controls, without occlusion augmentation and without pre-training), 12 in iteration 2 (V1–V4), 6 with YOLO26s, 6 in H6, 12 in H7 (two per rung of 2,000, 5,000 and 10,000 with seed 0, and six at 20,000), 6 in H8 and 3 in H9 (its V1 is that of iteration 2): 59 in the first seven blocks; then 6 in H9b, 6 in H10 (its V1 is that of H9b), 6 in H11, 9 in H12, 6 in the seed extension, 6 in H14, 6 in H15, 6 in H16 (V3 with diffusion and with depth; its V1 is that of H15), 6 in H17 and 1 replicate of the H15 V1 seed 0 on another machine: 117 in total. Versions: Ultralytics 8.4.143; PyTorch 2.11 with CUDA 12.8 on the laptop and 2.8 with CUDA 12.8 on RunPod; torchvision 0.23 for Faster R-CNN and RetinaNet. The V1 replicate differs from the original by −0.87 (NightOwls), −1.92 (CityPersons) and +3.42 pp (ExDark, 101 instances), above the 2 pp threshold declared in the pre-registration: it limits the H16 comparison, whose V1 are those of H15, trained on another machine (in the other blocks V1 and V3 share a machine, including H12, trained on a single pod).

Table S3: Run log by block (per-run detail in `informes/analisis_final/registro_corridas.csv`). Epochs: trained and of the evaluated weights, min–max. With validation selection, the evaluated epoch is the one with maximum validation mAP50-95 (ExDark val and occluded val).

Block	Detector	Runs	V1 ep. (eval./train.)	V3 ep. (eval./train.)	Exposure rule	Machine	Hours
Exp. 1 (parches)	YOLOv11s, YOLOv11s+CBAM	14	2 / 22–40	2 / 22	validation selection	RTX 5050 Laptop	7.8
Iteración 2 (siluetas)	YOLOv11s, YOLOv11s+CBAM	12	1–2 / 21–22	37–89 / 57–100	validation selection	RTX 5050 Laptop	10.0
Réplica YOLO26s	YOLO26s	6	1–3 / 21–23	82–87 / 100	validation selection	RTX 5050 Laptop	8.2
H6 (CityPersons)	YOLOv11s	6	38–51 / 58–71	85–88 / 100	validation selection	RTX 5050 Laptop	18.4
H7 peldaño 2 000	YOLOv11s	2	2 / 22	43 / 63	fixed presentations, patience 1/5	RTX 5050 Laptop	2.5
H7 peldaño 5 000	YOLOv11s	2	1 / 9	35 / 40	fixed presentations, patience 1/5	RTX 5050 Laptop	3.5
H7 peldaño 10 000	YOLOv11s	2	2 / 6	20 / 20	fixed presentations, patience 1/5	RTX 5050 Laptop	3.7
H7 peldaño 20 000	YOLOv11s	6	1–2 / 3–4	2–9 / 4–10	fixed presentations, patience 1/5	RTX 5050 Laptop	9.0
H8 (RT-DETR-l)	RT-DETR-l	6	—	—	validation selection	RTX 5050 Laptop	15.0
H9 (generativa)	YOLOv11s	3	—	1–2 / 21–22	validation selection	RTX 5050 Laptop	1.2
H9b (generativa, 30 ep.)	YOLOv11s	6	30 / 30	30 / 30	30 fixed epochs	RTX 5050 Laptop	3.5
H10 (geometría, 30 ep.)	YOLOv11s	3	—	30 / 30	30 fixed epochs	RTX 5050 Laptop	1.9
H10-s (siluetas, 30 ep.)	YOLOv11s	3	—	30 / 30	30 fixed epochs	RTX 5050 Laptop	1.8
H11 (CityPersons, 60 ep.)	YOLOv11s	6	60 / 60	60 / 60	60 fixed epochs	RTX 4090	1.9
H12 (YOLOv11m)	YOLOv11m	9	30 / 30	30 / 30	30 fixed epochs	RTX 4090	1.0
Extensión de semillas	YOLOv11s	6	30 / 30	30 / 30	30 fixed epochs	RTX 4090	0.5
H14 (profundidad, 30 ep.)	YOLOv11s	6	30 / 30	30 / 30	30 fixed epochs	RTX 4090	0.5
H15 (Faster R-CNN, 30 ep.)	Faster R-CNN R50-FPN v2	6	30 / 30	30 / 30	30 fixed epochs	RTX 4090	2.6
H16a (Faster R-CNN, generativa)	Faster R-CNN R50-FPN v2	3	—	30 / 30	30 fixed epochs	RTX 4090	1.9
H16b (Faster R-CNN, profundidad)	Faster R-CNN R50-FPN v2	3	—	30 / 30	30 fixed epochs	RTX 4090	1.9
H17 (RetinaNet, 30 ep.)	RetinaNet R50-FPN v2	6	30 / 30	30 / 30	30 fixed epochs	RTX 4090	2.5
Réplica de V1 entre pods	Faster R-CNN R50-FPN v2	1	30 / 30	—	30 fixed epochs	RTX 4090	0.2
Total		117					99.3

Table S4: Training recipe by detector. The base augmentation (identical in V1 and V3) uses the Ultralytics 8.4 values fixed in `config/aumento_base.yaml`: HSV (0.015, 0.7, 0.4), translation 0.1, scale 0.5, horizontal flip 0.5, mosaic 1.0 with no mosaic in the last 10 epochs.

Detector	Optimizer	LR (final)	Warm-up	Weight decay	Batch	Epochs
YOLOv11s/m, YOLO26s	SGD, momentum 0.937	0.002 (linear to 1 %)	3 epochs	$5 \cdot 10^{-4}$	16	up to 100 (pat. 20) or 30/60 fixed
RT-DETR-l	AdamW, β_1 0.9	10^{-4} (linear to 1 %)	3 epochs, bias 0	10^{-4}	8 (acc. to 64)	up to 100 (pat. 20)
Faster R-CNN R50-FPN v2	SGD, momentum 0.9	0.01, $\times$0.1 at 20 and 27	500 iterations	10^{-4}	8	30 fixed
RetinaNet R50-FPN v2	SGD, momentum 0.9	0.005, $\times$0.1 at 20 and 27	500 iterations	10^{-4}	8	30 fixed

4 Augmentation and control of each family

Table S5: Augmentation (training) and control (test) parameters. The control uses the test images of the training domain and all their valid people (the 841 ExDark test images; CityPersons val in H6 and H11, `cp_silueta_severa`), with coverage 0.70–1.00 and occluders the model did not see in training.

Family	Training (V3)	Control
Patches	Texture rectangles (0.4), object cut-outs (0.5) or opaque (0.1); $p = 0.5$ per image, 1–3 people, coverage 0.30–0.85; *train* cut-out bank	`opaco_severa`: opaque border band; *test* cut-out bank, disjoint from *train*
Silhouettes	Masked silhouettes of real objects and people (person with prob. 0.5), luminance matching; same p, number and coverage; `siluetas_train` bank	`silueta_severa`: `siluetas_test` bank (disjoint images)
Diffusion	Stable Diffusion 1.5 inpainting, 20 steps, guidance 7.5, ten COCO objects ("a parked car", "a bicycle"…) with the suffix "at night, low light, realistic photo"; two versions per image generated in advance and substituted with $p = 0.15$ (with mosaic) or 0.5 (without); quality check with YOLOv11s-seg (rejected if $>$30 % of the mask is still a person)	`generativa_severa`: same prompts and model, generated anew on the test images with other seeds (96.2 % accepted)
Real geometry	Silhouettes placed over the hidden region of real CityPersons *train* visibility patterns, coverage 0.40–0.95	`geometria_severa`: internal-validation patterns and *test* bank (mean IoU cut-out–hidden region 0.61)
Depth order	Object cut-outs standing on the ground in front of the person and composited only where the scene is farther (Depth Anything V2 Small); 2 versions per image, same substitution as diffusion	`profundidad_severa`: `siluetas_test` bank

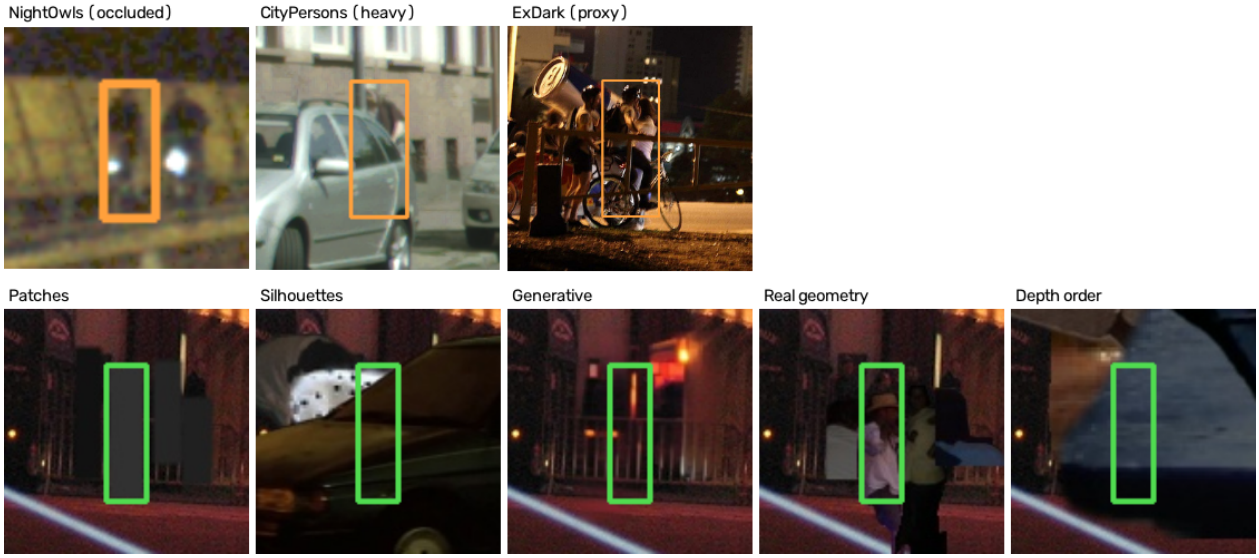


Figure S1: Top: annotated real occlusion in the three sources (orange box: evaluated person). Bottom: the same ExDark test person in each synthetic family's control (green box).

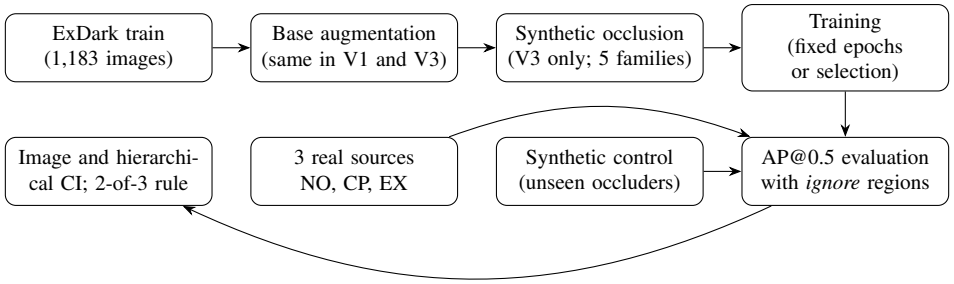


Figure S2: Pipeline of each block. Occlusion is applied to the RGB image after base augmentation and before formatting, in training only; validation and test are never occluded except in the control.

5 All cells and statistical analysis

Table S6: All cells: V3−V1 in AP@0.5 (pp), 95 % CI over images (1,000 replicates) and hierarchical (seeds and images, 2,000), 3 seeds per variant. * passes: estimate $\geq +2$ and image CI > 0. † rules out a $+2$ pp gain (upper image-CI limit $< +2$). Exposure: equal (fixed epochs), unequal (V1 and V3 selected at different epochs by validation) or early (both at epochs 1–2).

Block	Exp.	Cell	Δ	Image CI	Hierarchical CI	Cell	Δ	Image CI	Hierarchical CI
H3	early	Control	+60.92*	[+58.50; +63.41]	[+56.75; +64.95]	NO	−3.17†	[−4.58; −1.74]	[−7.94; +2.10]
		CP	−0.36†	[−1.06; +0.31]	[−2.42; +1.62]	EX	−3.25†	[−5.81; −0.21]	[−7.11; +0.89]
H4	unequal	Control	+10.63*	[+9.42; +12.15]	[+8.95; +12.50]	NO	−16.79†	[−19.59; −13.96]	[−20.41; −12.96]
		CP	−3.94†	[−5.66; −2.41]	[−7.14; −1.02]	EX	−5.02†	[−10.72; +0.50]	[−11.89; +1.28]
H4-Y26	unequal	Control	+12.01*	[+10.61; +13.57]	[+10.32; +13.79]	NO	−14.31†	[−17.35; −11.10]	[−18.07; −10.14]
		CP	−4.42†	[−5.95; −2.63]	[−6.89; −1.65]	EX	−3.51†	[−8.38; +1.21]	[−9.48; +1.86]
H6	unequal	Control	+23.74*	[+21.09; +26.51]	[+20.44; +27.22]	NO	−1.35†	[−2.85; +0.20]	[−3.47; +0.68]
		CP	+1.82	[+0.31; +3.31]	[−0.86; +4.42]	EX	+0.94	[−2.53; +4.72]	[−4.06; +6.68]
H7	unequal	NO	−1.43†	[−3.98; +1.00]	[−4.45; +1.56]	CP	−3.94†	[−5.62; −2.27]	[−8.15; +0.78]
		EX	−9.15†	[−13.94; −5.66]	[−17.22; −3.26]				
H8	unequal	Control	+10.97*	[+9.57; +12.36]	[+9.32; +12.83]	NO	−6.98†	[−8.95; −5.11]	[−11.16; −2.81]
		CP	−3.47†	[−4.86; −1.81]	[−6.27; −0.74]	EX	−2.21†	[−6.16; +1.59]	[−7.61; +3.39]
H9	early	Control	+2.60*	[+2.12; +3.09]	[+0.25; +4.33]	NO	−1.93†	[−2.89; −0.88]	[−3.87; −0.12]
		CP	+1.23	[+0.36; +2.11]	[−1.70; +4.03]	EX	+2.82*	[+0.37; +5.99]	[−1.23; +7.40]
H9b	equal	Control	+15.23*	[+14.00; +17.16]	[+13.81; +17.45]	NO	+0.01†	[−1.02; +1.20]	[−1.60; +1.82]
		CP	+1.52	[+0.79; +2.66]	[−0.02; +3.39]	EX	−3.03†	[−6.77; +0.99]	[−8.85; +2.89]
H11	equal	Control	+24.66*	[+21.62; +27.47]	[+21.44; +28.09]	NO	+1.95	[+0.93; +3.21]	[−0.16; +4.66]
		CP	+1.31	[−0.28; +2.78]	[−0.85; +3.46]	EX	−2.44	[−6.77; +2.36]	[−7.34; +3.23]
H10	equal	Control	+18.39*	[+15.61; +21.30]	[+15.57; +21.47]	NO	−2.05†	[−3.26; −0.92]	[−4.62; +0.29]
		CP	+2.37*	[+1.52; +3.52]	[+0.63; +4.61]	EX	−4.88†	[−9.08; −0.82]	[−11.58; +1.03]
H10-s	equal	Control	+10.99*	[+9.62; +12.62]	[+9.58; +12.68]	NO	−1.12†	[−2.13; +0.08]	[−2.58; +0.49]
		CP	+2.26*	[+1.44; +3.47]	[+0.72; +4.04]	EX	−3.50†	[−7.64; +0.08]	[−9.63; +1.86]
H12a	equal	Control	+14.45*	[+12.76; +16.37]	[+12.67; +16.46]	NO	−0.39†	[−1.55; +0.89]	[−3.92; +2.77]
		CP	+0.25†	[−0.63; +1.27]	[−1.32; +2.12]	EX	+3.04	[−1.07; +6.63]	[−2.63; +8.31]
H12b	equal	Control	+19.04*	[+17.49; +21.03]	[+17.35; +21.21]	NO	+0.78	[−0.42; +2.16]	[−3.21; +4.28]
		CP	+1.11	[+0.14; +2.16]	[−0.47; +2.89]	EX	−0.56	[−4.07; +3.19]	[−5.77; +5.06]
H14	equal	Control	+7.42*	[+6.40; +8.79]	[+6.13; +9.13]	NO	−1.50†	[−2.75; −0.37]	[−3.40; +0.26]
		CP	−0.16†	[−1.21; +0.75]	[−1.85; +1.31]	EX	+5.49*	[+1.63; +9.30]	[−0.35; +11.14]
H15	equal	Control	+14.72*	[+13.04; +16.66]	[+12.97; +16.63]	NO	+3.28*	[+1.64; +4.79]	[+0.67; +5.81]
		CP	+2.33*	[+1.01; +3.69]	[+0.13; +4.92]	EX	−1.00	[−5.61; +3.24]	[−6.71; +4.04]
H16a	equal	Control	+20.35*	[+18.57; +22.33]	[+18.53; +22.26]	NO	+0.92	[−0.54; +2.26]	[−1.65; +3.58]
		CP	−0.76†	[−1.84; +0.50]	[−2.58; +1.16]	EX	+0.80	[−3.54; +5.02]	[−5.12; +6.34]
H16b	equal	Control	+9.63*	[+8.36; +11.34]	[+8.17; +11.57]	NO	+0.37†	[−1.00; +1.66]	[−2.12; +2.61]
		CP	+0.60	[−0.71; +2.05]	[−1.58; +2.79]	EX	−0.45	[−6.19; +4.64]	[−6.72; +5.63]
H17	equal	Control	+6.60*	[+5.61; +7.81]	[+5.23; +8.28]	NO	−2.11†	[−3.53; −0.67]	[−7.25; +2.77]
		CP	−0.28†	[−1.16; +0.73]	[−2.14; +1.50]	EX	−3.77†	[−8.47; +0.82]	[−9.89; +2.78]

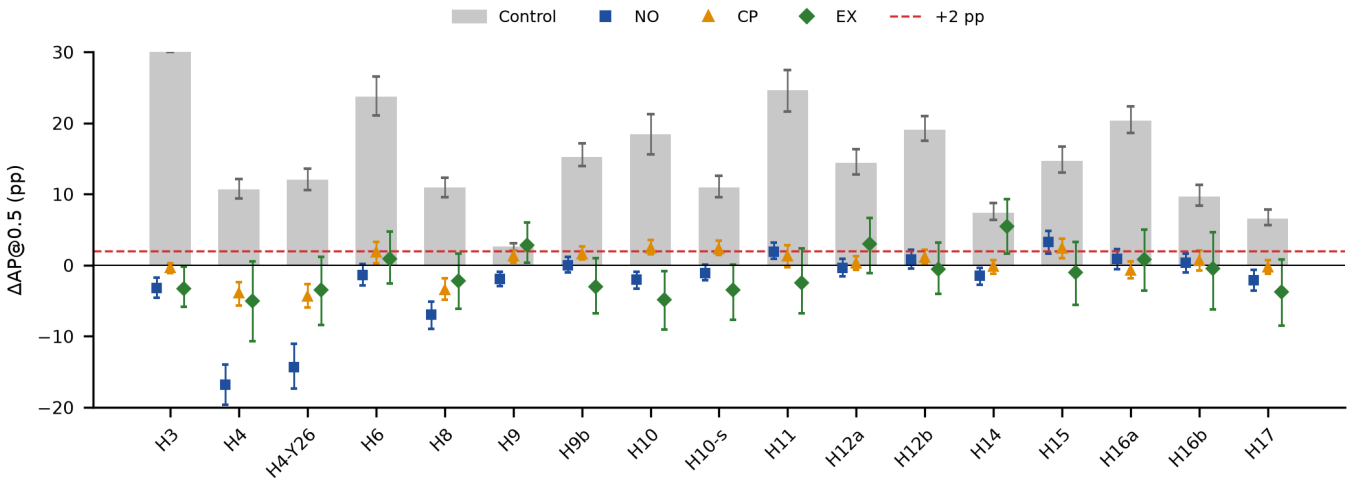


Figure S3: Synthetic control (bars) and real sources (markers) per block, with 95 % image CIs. The H3 bar is cut at +30 (+60.9).

Table S7: Counts and declared (post hoc) statistical analyses: 13 blocks with a control, 39 real cells. Mixed model: real gain $\sim$ synthetic gain (standardized) + source, with a random intercept per block.

Real cells that pass (estimate $\geq +2$ and CI > 0)	4 of 39 (hierarchical CI: 2)
Rule out a +2 pp gain (non-superiority, one-sided 2.5 %)	24 of 39 (hierarchical: 19)
Equal exposure (21 cells): pass / negative with CI / mean	3 / 3 / +0.02 pp
Unequal exposure (12 cells): pass / negative with CI / mean	0 / 6 / -4.94 pp
Early selection in both (6 cells): pass / negative with CI / mean	1 / 3 / -0.78 pp
Without the ExDark proxy: passing cells; blocks with 2 of 2	2 of 26; 0
Between-seed SD of V1 AP@0.5 on real sources (median; max.)	1.53; 4.10 pp (26 groups)
Spearman ρ, NO (block-bootstrap CI)	+0.41 [-0.18; +0.85]
Spearman ρ, CP (block-bootstrap CI)	+0.30 [-0.26; +0.71]
Spearman ρ, EX (block-bootstrap CI)	-0.15 [-0.67; +0.54]
Mixed model: slope per SD of synthetic gain (14.0 pp)	+0.11 pp [-1.74; +1.95], $p = 0.91$
Same without H3	+0.66 pp [-1.31; +2.64], $p = 0.51$

Why +2 pp (review of 22 October). Besides exceeding the between-seed SD of V1 AP (median 1.5 pp), +2 pp is practically relevant: on the median V1 AP@0.5 for occluded people (0.17 on NightOwls, 0.15 on CityPersons *heavy* and 0.26 on the ExDark proxy) it is a relative improvement of 12 %, 13 % and 8 %. And it is detectable: the median standard error of a V3$-$V1 difference is 0.78 pp (0.68 on NightOwls, 0.52 on CityPersons and 1.98 on ExDark), so, except on the ExDark proxy, a +2 pp effect lies more than 2.5 standard errors from zero.

6 Cross-family control matrix

If the control measured only memory of the specific occluder, a model would gain on its own control and not on those of other families. Table S8 evaluates every family on every control. 13 of the 20 off-diagonal cells pass (also 13 with the hierarchical CI): augmentation transfers from one synthetic occlusion to another, but not to real occlusion (primary count: 5 of 33). The experiment-1 patches and diffusion stay close to their own control.

Table S8: Cross-family control matrix: V3$-$V1 (AP@0.5, pp, 3 seeds) of each training family (rows) on each family's severe control (columns), image CI in brackets. All controls use ExDark test images and occluder instances unseen in training. Patches: experiment-1 models (epochs 1–2); the rest, 30 fixed epochs.

Trained with	Patches	Silhouettes	Generative	Real geometry	Depth order
Patches	**+60.9** [+58.52; +63.35]	+1.3 [+0.79; +1.86]	+3.0 [+2.54; +3.71]	+0.9 [+0.15; +1.75]	+0.8 [+0.49; +1.21]
Silhouettes	+18.1 [+16.56; +19.99]	**+11.0** [+9.63; +12.57]	+2.3 [+1.84; +3.14]	+15.8 [+13.48; +18.61]	+4.3 [+3.55; +5.07]
Generative	+7.6 [+6.71; +8.63]	+0.4 [+0.02; +0.87]	**+15.2** [+14.01; +17.20]	+1.9 [+0.97; +2.97]	+0.7 [+0.31; +1.05]
Real geometry	+13.7 [+12.37; +15.26]	+6.3 [+5.19; +7.48]	+1.7 [+1.28; +2.55]	**+18.4** [+15.77; +21.26]	+2.2 [+1.64; +2.79]
Depth order	+8.3 [+7.32; +9.60]	+6.7 [+5.66; +7.95]	+3.1 [+2.40; +3.72]	+11.7 [+9.67; +14.52]	**+7.4** [+6.35; +8.79]

7 Secondary analyses

Table S9: Declared secondary analyses: five seeds (YOLOv11s, 30 epochs), the H7 NightOwls ladder (seed 0, NightOwls occluded) and miss rate MR^{-2} (pp; negative = augmentation misses fewer people).

Analysis	Cell	Δ	CI (images / hierarchical)
H9b, 5 seeds	NO	−0.45	[−1.28; +0.60] / [−1.99; +1.23]
H9b, 5 seeds	CP	+0.22	[−0.37; +1.02] / [−1.39; +2.00]
H9b, 5 seeds	EX	−1.75	[−4.79; +1.83] / [−6.80; +3.46]
H10-s, 5 seeds	NO	−1.06	[−1.89; −0.03] / [−2.36; +0.39]
H10-s, 5 seeds	CP	+2.06	[+1.40; +2.99] / [+0.26; +4.17]
H10-s, 5 seeds	EX	−2.06	[−5.37; +0.97] / [−7.27; +2.65]
H7, 2000 img.	NO	+0.66	[−3.31; +5.13]
H7, 5000 img.	NO	−1.12	[−4.99; +3.10]
H7, 10000 img.	NO	+4.38	[+0.17; +8.06]
H7, 20000 img.	NO	−1.43	[−3.98; +1.00]
MR^{-2} H3	NO +2.9; CP −0.0; EX +2.4		
MR^{-2} H4	NO +13.4; CP +1.6; EX +8.2		
MR^{-2} H4-Y26	NO +10.3; CP +1.6; EX +3.4		
MR^{-2} H6	NO +2.1; CP −3.2; EX +0.2 (1 improve ≥2 with CI)		
MR^{-2} H7	NO +1.4; CP +2.4; EX +14.7		
MR^{-2} H8	NO +6.3; CP +2.1; EX +2.8		
MR^{-2} H9	NO +1.4; CP −0.9; EX −3.2 (1 improve ≥2 with CI)		
MR^{-2} H9b	NO −0.2; CP −1.2; EX +2.9		
MR^{-2} H10	NO +1.7; CP −1.5; EX +4.1		
MR^{-2} H10-s	NO +0.8; CP −1.5; EX +3.0		
MR^{-2} H11	NO −1.8; CP −0.6; EX +1.1		
MR^{-2} H12a	NO +0.5; CP +0.0; EX −4.0 (1 improve ≥2 with CI)		
MR^{-2} H12b	NO +0.3; CP −0.5; EX +0.6		
MR^{-2} H14	NO +0.9; CP +0.5; EX −3.4 (1 improve ≥2 with CI)		
MR^{-2} H15	NO −2.0; CP −1.4; EX −1.3		
MR^{-2} H16a	NO +0.1; CP +0.0; EX +0.3		
MR^{-2} H16b	NO −0.1; CP −0.5; EX +0.8		
MR^{-2} H17	NO +2.3; CP +0.2; EX +4.4		

Table S10: CityPersons by annotated visibility (declared post hoc analysis): V3−V1 in AP@0.5 (pp) per bin; other people are *ignore* regions.

Block	0.20–0.35	0.35–0.50	0.50–0.65	0.65–0.80	0.80–0.90	0.90–1.00
H3	−0.1	+0.4	+0.3	+1.2	+1.1	+1.7
H4	−0.1	−1.8	−5.4	−9.7	−13.8	−11.9
H4-Y26	−0.3	−1.9	−5.8	−8.5	−12.8	−9.9
H6	+4.3	+3.4	+2.1	+0.4	+0.8	+0.4
H7	+0.3	−1.3	−8.1	−14.1	−16.4	−15.6
H8	−0.1	−2.5	−6.3	−9.2	−9.8	−7.2
H9	+0.3	+1.0	+1.0	+2.5	+3.0	+2.1
H9b	+0.0	+0.8	+2.6	+3.9	+5.9	+5.2

Foundation-model reference. OWLv2 (Minderer et al., NeurIPS 2023) and Grounding DINO (Liu et al., ECCV 2024) without fine-tuning do not beat the COCO detectors on occluded NightOwls or CityPersons *heavy*, which training on CityPersons takes to 0.58–0.59 (`informes/reporte_referencia_fundacion.md`).

Realism (H13). Mean coverage achieved: patches 0.56, silhouettes 0.54, geometry 0.47 and diffusion 0.32 (the quality check left 146 of 400 people unoccluded), against 0.55 for real occlusion. CityPersons has no real occluder masks (only the visible and full boxes), so the IoU between synthetic and real masks cannot be computed; the real-geometry family reproduces the hidden region of real patterns by construction (mean IoU 0.61 between cut-out and hidden region) and does not transfer either (H10).

KID on the occluder region (post hoc). With DINOv2 on the occluded part only (real: strip of the full box outside the annotated visible box; synthetic: rectangle of the pixels that change with respect to the same unoccluded person), the distance to real occluders is 2.56 [2.08; 3.17] for patches, 3.39 [2.63; 4.23] for silhouettes, 2.75 [1.91; 3.51] for geometry and 2.50 [1.87; 3.02] for diffusion (248 regions; the quality check left 152 unoccluded), against 3.63 [2.77; 4.34] for the same region without an occluder and 0.01 [−0.08; 0.10] between two halves of real occluders. Synthetic occluders are somewhat closer to real occluders than an unoccluded region, but none comes close (`informes/kid_oclusor.py`).

8 Post hoc analyses requested by the review of 22 October

These analyses were declared after seeing the results and change no pre-registered rule or verdict. Specificity compares, within the same source, the change on occluded people with the change on unoccluded people; a positive value can arise because visible

people get worse, so it is read together with the change on occluded people (Table I of the paper). The visible box separates seeing a partly hidden person from estimating its full extent. The V1 replicate on another machine (a single seed) only bounds the variation between machines.

Table S11: Specificity and box convention (post hoc analyses, review of 22 October). Specificity: [V3−V1] on occluded people minus [V3−V1] on unoccluded people of the same source (NO: occluded − none; CP: *heavy* − *bare*; EX: severe − none), with the same images resampled for both strata (1,000 replicates). * CI > 0 (the gain is larger for occluded people); $^-$ CI < 0. CP *heavy*: V3−V1 scored with the full box (CityPersons protocol) and with the visible box (`bboxVis` from `anno_val.mat`), IoU ≥ 0.5.

Block	Specificity NO	Specificity CP	Specificity EX	CP full box	CP visible box
H3	−2.15$^-$ [−3.34; −0.94]	−2.02$^-$ [−2.95; −1.06]	−1.92 [−4.56; +0.88]	−0.36	+0.00 [−0.44; +0.47]
H4	+1.62 [−1.10; +4.28]	+8.00* [+5.28; +10.89]	−3.06 [−8.41; +1.99]	−3.94	−4.12 [−5.74; −2.99]
H4-Y26	−1.56 [−4.42; +1.17]	+5.49* [+3.20; +7.95]	−5.03$^-$ [−10.04; −0.61]	−4.42	−3.92 [−4.96; −2.72]
H6	+3.92* [+2.45; +5.50]	+1.41 [−0.19; +2.93]	+0.39 [−3.33; +4.11]	+1.82	+0.70 [−0.56; +2.01]
H7	+0.35 [−2.06; +2.67]	+11.67* [+9.36; +14.05]	+8.53* [+3.54; +12.51]	−3.94	−4.61 [−5.97; −3.17]
H8	+1.23 [−0.46; +3.16]	+3.74* [+2.17; +5.44]	−0.97 [−4.69; +2.75]	−3.47	−3.81 [−4.93; −2.69]
H9	−0.79 [−1.67; +0.09]	−0.85 [−2.00; +0.19]	+2.76* [+0.31; +5.81]	+1.23	−0.20 [−0.81; +0.44]
H9b	−1.17 [−2.23; +0.13]	−3.72$^-$ [−5.13; −2.24]	−2.03 [−5.74; +1.63]	+1.52	+0.29 [−0.13; +1.10]
H10	−2.11$^-$ [−3.17; −0.95]	−2.88$^-$ [−4.28; −1.31]	−5.36$^-$ [−9.57; −1.24]	+2.37	+0.05 [−0.35; +0.70]
H10-s	−2.94$^-$ [−3.75; −1.65]	−3.71$^-$ [−5.08; −2.20]	−3.35 [−7.80; +0.35]	+2.26	+0.63 [+0.11; +1.37]
H11	−1.39 [−2.63; +0.00]	+0.59 [−0.97; +2.21]	−10.08$^-$ [−14.18; −5.42]	+1.31	+0.99 [−0.02; +1.77]
H12a	−0.22 [−1.24; +1.06]	−0.78 [−1.98; +0.54]	+2.70 [−1.56; +6.60]	+0.25	+0.37 [−0.25; +0.79]
H12b	−1.86$^-$ [−3.10; −0.42]	−3.84$^-$ [−5.29; −2.31]	+0.01 [−3.61; +3.45]	+1.11	+1.25 [+0.55; +1.70]
H14	−2.57$^-$ [−3.73; −1.42]	−0.98 [−2.46; +0.24]	+5.27* [+1.67; +8.69]	−0.16	−0.22 [−0.84; +0.33]
H15	−1.09 [−2.49; +0.23]	+1.45 [−0.04; +3.08]	−0.69 [−5.33; +3.55]	+2.33	+0.74 [+0.04; +1.41]
H16a	+2.85* [+1.54; +4.10]	+2.58* [+1.24; +4.16]	+1.53 [−3.01; +5.47]	−0.76	−0.76 [−1.69; +0.04]
H16b	+0.06 [−1.14; +1.17]	+2.26* [+0.63; +3.76]	+0.18 [−5.02; +5.16]	+0.60	−0.14 [−0.98; +0.73]
H17	+4.60* [+3.26; +6.02]	+4.20* [+2.83; +5.65]	+3.07 [−1.33; +7.60]	−0.28	−0.63 [−1.48; +0.13]

Table S12: Primary count and statistical analyses requested by the review (post hoc; `informes/revision_panel_analisis.py`). In brackets, with the hierarchical CI.

Primary count (equal exposure: 11 blocks, 33 cells): pass / rule out +2 / inconclusive	5 (4) / 15 (9) / 13
Mean NO with equal exposure, block-bootstrap CI; positive blocks	+0.01 [−0.92; +0.97]; 6 of 11
Mean CP with equal exposure, block-bootstrap CI; positive blocks	+0.96 [+0.34; +1.56]; 8 of 11
Mean EX with equal exposure, block-bootstrap CI; positive blocks	−0.94 [−2.55; +0.94]; 3 of 11
Random effects NO: mean, 90 % CI (TOST ±2 pp), τ	−0.01 [−0.80; +0.77], τ=1.45; equivalent
Random effects CP: mean, 90 % CI (TOST ±2 pp), τ	+0.95 [+0.38; +1.51], τ=0.98; equivalent
Random effects EX: mean, 90 % CI (TOST ±2 pp), τ	−0.88 [−2.50; +0.75], τ=2.49; not equivalent
Power of the "2 of 3" rule: P(detect) with a true +2 pp on all three sources (median [range]); with +3 pp	0.33 [0.30; 0.34]; 0.94
Multiplicity (51 cells, independence assumed): expected passes under the null, image / hierarchical error; observed	0.56 / 1.13; 6
P(some block reaches "2 of 3") by chance), image / hierarchical error	0.004 / 0.025
CI with between-seed variance (Welch's t, df 3.2): H15 NO	+3.28 [−0.14; +6.71]
CI with between-seed variance (Welch's t, df 3.7): H15 CP	+2.33 [−0.86; +5.52]
CI with between-seed variance (Welch's t, df 2.9): H15 EX	−1.00 [−3.48; +1.48]
CI with between-seed variance (Welch's t, df 2.2): H9 EX	+2.82 [−3.22; +8.87]
CI with between-seed variance (Welch's t, df 3.1): H10 CP	+2.37 [−0.55; +5.28]
CI with between-seed variance (Welch's t, df 3.9): H10-s CP	+2.26 [+0.42; +4.11]
H16a NO: with the H15 V1 / with the V1 replicate (other machine, one seed)	+0.92 / +0.76
H16b NO: with the H15 V1 / with the V1 replicate (other machine, one seed)	+0.37 / +0.21
H16a CP: with the H15 V1 / with the V1 replicate (other machine, one seed)	−0.76 / −0.14
H16b CP: with the H15 V1 / with the V1 replicate (other machine, one seed)	+0.60 / +1.22
H16a EX: with the H15 V1 / with the V1 replicate (other machine, one seed)	+0.80 / −2.55
H16b EX: with the H15 V1 / with the V1 replicate (other machine, one seed)	−0.45 / −3.80
Spearman ρ control–real, 17 blocks (NO; CP; EX)	+0.37 [−0.11; +0.75]; +0.24 [−0.21; +0.62]; −0.05 [−0.55; +0.51]
Spearman ρ control–real, equal exposure (NO; CP; EX)	+0.63 [+0.00; +1.00]; +0.25 [−0.46; +0.89]; −0.05 [−0.76; +0.64]

Table S13: Cross-family control matrix of the torchvision detectors: V3−V1 (pp) of each block on each family's control; in bold, its own control.

Block	Silhouettes	Diffusion	Geometry	Depth
H15 Faster R-CNN, silhouettes	**+14.7**	+2.9	+20.5	+7.8
H16a Faster R-CNN, diffusion	+0.5	**+20.4**	+1.4	+0.6
H16b Faster R-CNN, depth	+8.6	+2.7	+14.1	**+9.6**
H17 RetinaNet, silhouettes	**+6.6**	+1.8	+8.2	+3.2

Table S14: Additional controls with the same weights of each block (post hoc analyses; `informes/controles_extra.py`). Synthetic CP: silhouettes pasted on CityPersons val images (`cp_silueta_severa`, coverage 0.70–1.00), evaluated with models trained on ExDark: it separates the domain shift from occlusion. Moderate: silhouettes on ExDark test with coverage 0.35–0.70, similar to CityPersons *heavy*. V3−V1 (pp) with image CI; * passes. On the right, real CityPersons *heavy* (Table I of the paper).

Block	Synthetic CP	Moderate (ExDark)	Real CP *heavy*
H3	+0.18 [−0.22; +0.45]	+2.10* [+1.12; +3.05]	−0.36
H4	+3.01* [+2.05; +4.66]	+24.53* [+22.13; +26.63]	−3.94
H4-Y26	+5.51* [+4.02; +7.28]	+25.93* [+23.53; +28.19]	−4.42
H8	+5.59* [+4.39; +7.28]	+28.09* [+25.75; +30.41]	−3.47
H9	−0.11 [−0.62; +0.18]	+2.40* [+1.65; +3.11]	+1.23
H9b	+0.31 [−0.38; +0.65]	+7.06* [+5.66; +8.52]	+1.52
H10	+1.91 [+1.10; +2.83]	+21.99* [+19.86; +23.89]	+2.37
H10-s	+3.76* [+2.68; +5.02]	+25.38* [+23.16; +27.41]	+2.26
H12a	+8.33* [+6.76; +10.65]	+32.13* [+29.78; +34.56]	+0.25
H12b	+0.90 [+0.32; +1.67]	+8.68* [+7.16; +10.27]	+1.11
H14	+2.27* [+1.53; +3.16]	+17.53* [+15.39; +19.51]	−0.16
H15	+7.70* [+6.29; +9.39]	+33.55* [+31.23; +35.63]	+2.33
H16a	+0.39 [−0.09; +0.69]	+7.62* [+6.20; +9.10]	−0.76
H16b	+5.09* [+3.63; +6.58]	+20.57* [+18.56; +22.49]	+0.60
H17	+2.37* [+1.74; +3.35]	+11.58* [+9.88; +13.11]	−0.28

9 Literature search

Three Scopus searches exported on 13 Oct 2026: main equation (A and B, plus at least one of C, D or E; 155 records), two complementary equations (8) and synthetic versus real occlusion without restricting to people (71). They yielded the five seed articles (PRNet++, WALT, MLFFCSP, CamPed and the YOLOX with domain transfer); the search was supported by Scopus AI and Claude, and the authors made the selection by reading the articles. Exact strings, copied from the Scopus search history:

Main equation (155 records)

```
TITLE-ABS-KEY ( ( ( pedestrian* or person or persons or people or human or humans ) W/4 detect* ) AND ( occlu* OR "partially visible" OR "partial visibility" OR "crowded scene" ) AND ( ( occlu* W/5 ( augment* or synthetic* or simulat* or artificial* or generat* or inpaint* or diffusion or paste or pasted or pasting or erasing ) ) OR "random erasing" OR cutout OR "copy paste" OR "low light" OR "low illumination" OR "poor illumination" OR "poor lighting" OR "dim light" OR "dark scene" OR "dark environment" OR "dark image" OR darkness OR night* OR nocturnal OR exdark OR nightowls OR "cross-dataset" OR "cross-domain" OR "domain gap" OR "domain shift" OR "dataset bias" OR "domain generali?ation" OR "out-of-distribution" OR "sim-to-real" OR "synthetic-to-real" OR "real occlusion" OR "natural occlusion" OR "real-world occlusion" OR ( generali?ation W/5 ( dataset* or domain* ) ) ) ) AND PUBYEAR > 2021 AND PUBYEAR < 2027 AND ( DOCTYPE ( ar ) OR DOCTYPE ( cp ) OR DOCTYPE ( re ) ) AND LANGUAGE ( english ) AND ( subjarea ( COMP ) OR subjarea ( ENGI ) ) AND NOT TITLE ( thermal OR infrared OR multispectral OR lidar OR radar OR "point cloud" )
```

Complementary equation (8 records)

```
TITLE-ABS-KEY ( ( ( pedestrian* or person or persons or people or human or humans ) W/4 detect* ) AND ( ( occlu* W/5 ( augment* or synthetic* or simulat* or artificial* or generat* or inpaint* or diffusion or paste or pasted or pasting or erasing ) ) OR "random erasing" OR cutout OR "copy paste" ) AND ( "low light" OR "low illumination" OR "poor illumination" OR night* OR nocturnal OR exdark OR nightowls OR "cross-dataset" OR "cross-domain" OR "domain gap" OR "domain shift" OR "real occlusion" OR "natural occlusion" OR "real-world occlusion" ) ) AND ( DOCTYPE ( ar ) OR DOCTYPE ( cp ) OR DOCTYPE ( re ) ) AND LANGUAGE ( english )
```

Synthetic versus real occlusion (71 records)

```
TITLE-ABS-KEY ( ( occlu* W/5 ( synthetic* or simulat* or artificial* ) ) AND ( "real occlusion" OR "natural occlusion" OR "real-world occlusion" OR "naturally occluded" OR "real occluded" ) ) AND ( DOCTYPE ( ar ) OR DOCTYPE ( cp ) OR DOCTYPE ( re ) ) AND LANGUAGE ( english )
```

After two reviews (20 Oct 2026), Taori et al. (2020), Pytel et al. (2021), Kortylewski et al. (2020), Wang et al. (2020), Ghiasi et al. (2021) and Geirhos et al. (2020) were added by targeted search after reading their abstract or text. All references were verified in Crossref or arXiv.

10 Licences, data and compliance

Licences. ExDark: images under its terms of use for non-commercial academic research (`https://github.com/cs-chan/Exclusively-Dark-Image-Dataset`). CityPersons (Cityscapes images): non-commercial use, cite the dataset, no redistribution (`https://www.cityscapes-dataset.com/license/`). NightOwls: non-commercial use, cite the dataset, no redistribution; abstract derivatives such as trained models are allowed (`https://www.nightowls-dataset.org`). The project does not redistribute images; it publishes identifiers, derived annotations, code and weights.

Repository material. `PREREGISTRO_*.md`, `CAMBIOS_PROTOCOLO.md` and `protocolo_congelado.json` (hypotheses and fingerprints); `informes/analisis_final/registro_corridas.csv` (runs); `celdas_completas.`

csv, bootstrap*.csv, miss_rate*.csv and revision_sp.json (figures and intervals); config/ and argos_lib/ (augmentation and control); 16_oclusion_proxy.py and gt_test_proxy.json (ExDark proxy); 17_fiabilidad_anotacion.py (κ study, without annotator names); 31_realismo.py (H13).

Table S15: Summarised REFORMS checklist (Kapoor et al., 2024): where each block is addressed.

Block	Where
Objectives and claims	Introduction; Table I of the paper; Table S2
Computational reproducibility	Public repository https://github.com/jhanrq8888/argos-occlusion-transfer; weights with a DOI on Zenodo upon publication; versions in Section 3
Data and preprocessing	Data section of the paper; augmentation and control table; licences
Train–test separation	Control with unseen images and occluders; validation never occluded except val_ocluido for selection
Model selection	Table S3 (evaluated epoch of each run); fixed-exposure blocks as primary evidence
Metrics and uncertainty	Tables S6 and S7; MR^{-2} in Table S9; AP@0.5:0.95 in informes/analisis_final/ap_estricto.csv (none of the 21 cells of the first seven blocks passes)
Generalisation and limitations	Discussion of the paper (three categories)

Practical guide. (1) A synthetic occlusion benchmark is a positive control that an augmentation is correctly implemented, not evidence of robustness. (2) Claiming occlusion robustness requires a set of annotated real occlusion from the deployment domain, even a small one, compared with equal training exposure. (3) In a descriptive, uncontrolled comparison, training on annotated in-domain data gave far more than any augmentation (0.59 versus 0.24 on CityPersons *heavy*); comparing them at equal budget is pending. (4) Always report the epoch of the V1 and V3 weights.